CHAPTER 7: PRINCIPLES OF DIGITAL DATA TRANSMISSION

- This chapter deals with the problems of transmitting digital data over a communication channel.
- Hence, the starting messages are assumed to be digital.
- We shall begin by considering the binary case, where the data consist of only two symbols: **1** and **0**.
- We assign a distinct pulse p(t) to each one of these two symbols.
- The resulting sequence of these pulses is transmitted over the communication channel.
- At the receiver, these pulses are detected and are converted back to binary data **(1s** and **0s)**.

7.1 DIGITAL COMMUNICAION SYSTEMS

- A digital communication system consists of several components, as shown in Fig. 7. 1.

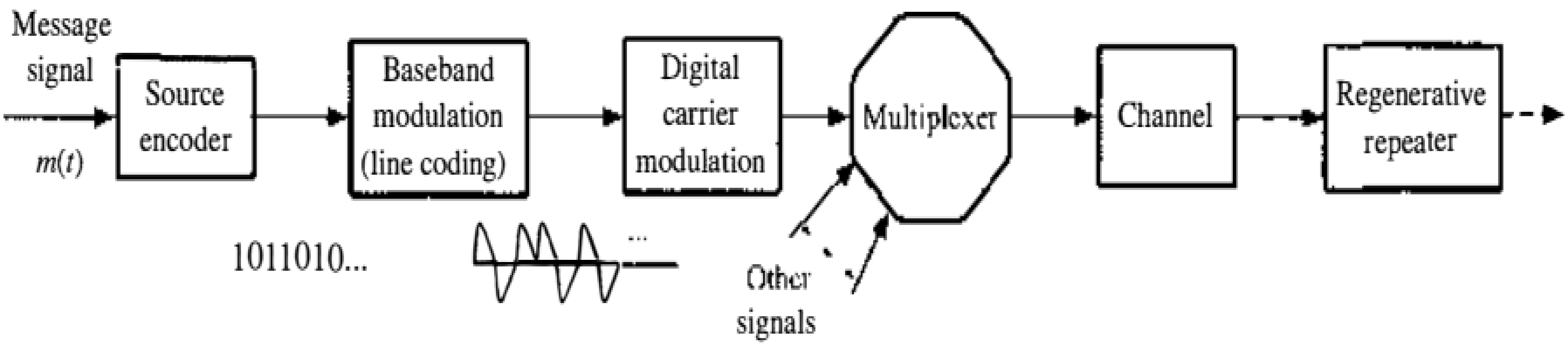


Figure 7. 1: Fundamental blocks of digital communication systems

- A digital message is nothing more than an **ordered sequence of symbols** produced by a discrete information source. The source produces output symbols at some average **symbol rate** R_s *symbols per second.*
- The finite transmission bandwidth of the channel sets an upper limit to the symbol rate R_b

7.1.1 Source

- The input to a digital system takes the form of a sequence of digits. The input could be the output from a data set, a computer, or a digitized audio signal (PCM, DM, or LPC), digital facsimile or HDTV, or telemetry data, and so on.

7.1.2 Line Codes

- Line coding is the process of converting the digital output of a source encoder into electrical pulses (waveforms) for the purpose of transmission over the channel.
- There are many possible ways of assigning pulses to the digital data.
- There are two types of codes according to the pulse width used, are three types of codes according to the polarity of the pulse used.

a. Line codes according to the pulse width

- Non Return to Zero (NRZ): The pulse amplitude is held to a constant value throughout the pulse interval (i.e., it does not have a chance to go to zero before the next pulse begins).
- Return to Zero (RZ): The pulse amplitude is held to a constant value throughout a portion of the pulse interval (e.g., half pulse interval) followed by a return to the zero level.

b. Line codes according to the polarity of the pulse

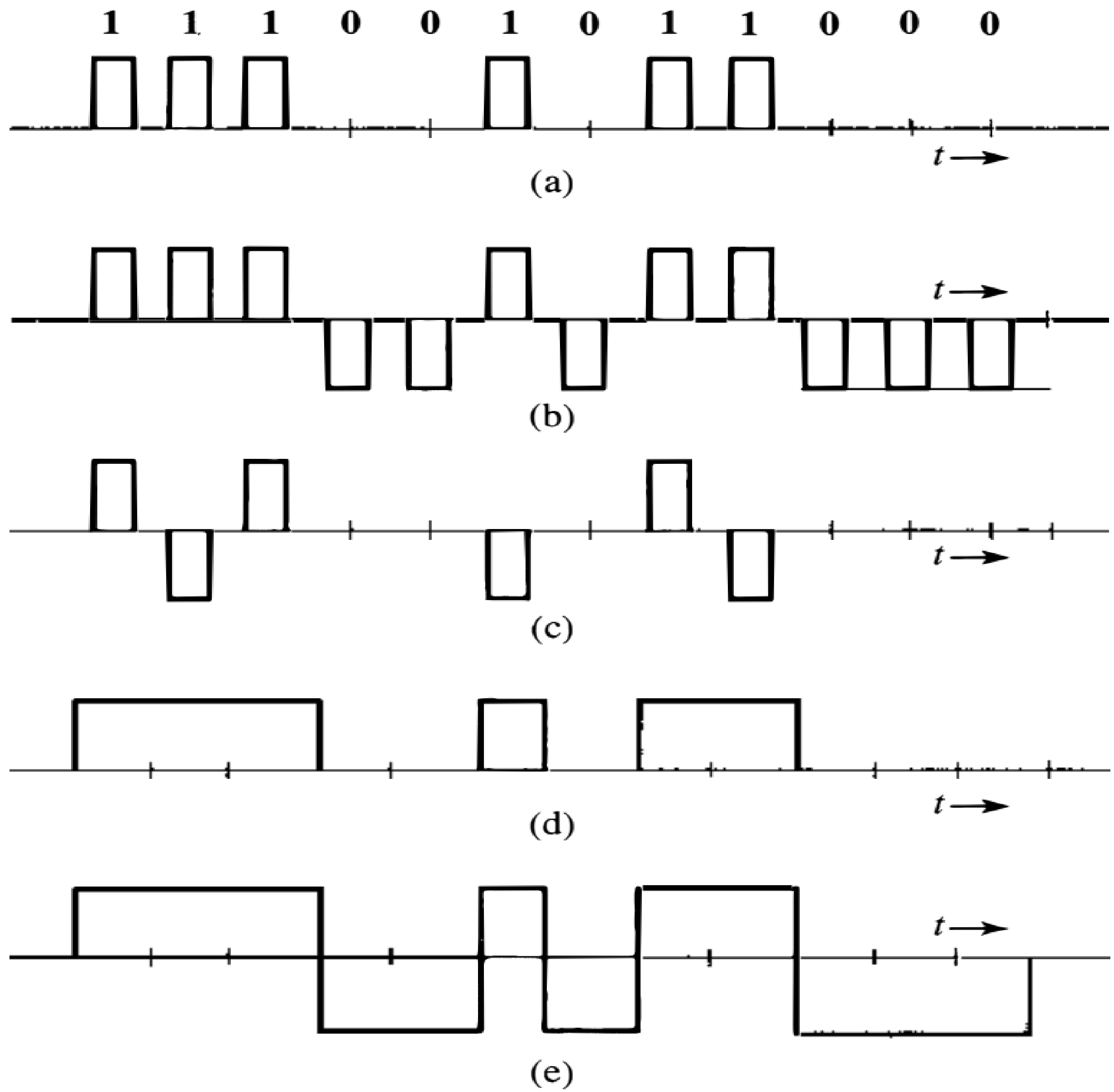
- Unipolar (on-off): a **1** is transmitted by a pulse *p(t)* and a **0** is transmitted by no pulse (zero signal).
- Polar: a **1** is transmitted by a positive pulse *p(t)* and **0** is transmitted by a negative pulse *-p(t).*

- Bipolar [*alternate mark inversion (AMI)*]: a **0** is encoded by no pulse and consecutive **1s** alterate in sign (i.e., if the current **1** is encoded by *p(t), the next* **1** *should be encoded by -p(t).*

Note: For every type of the above codes, it is possible to use RZ or NRZ coding as shown in Fig. 7.2

- The duobinary, and modified duobinary are better than the bipolar in terms of bandwidth efficiency.
- the modified duobinary line code, has seen applications in hard disk drive read channels, in optical 10 Gbit/s transmission for metronetworks, and in the first-generation modems for integrated services digital networks (ISDN). Details of duobinary line codes will be discussed later in this chapter.

Figure 7.2
Line code examples:
(a) on-off (RZ);
(b) polar (RZ);
(c) bipolar (RZ);
(d) on-off (NRZ);
(e) polar (NRZ).



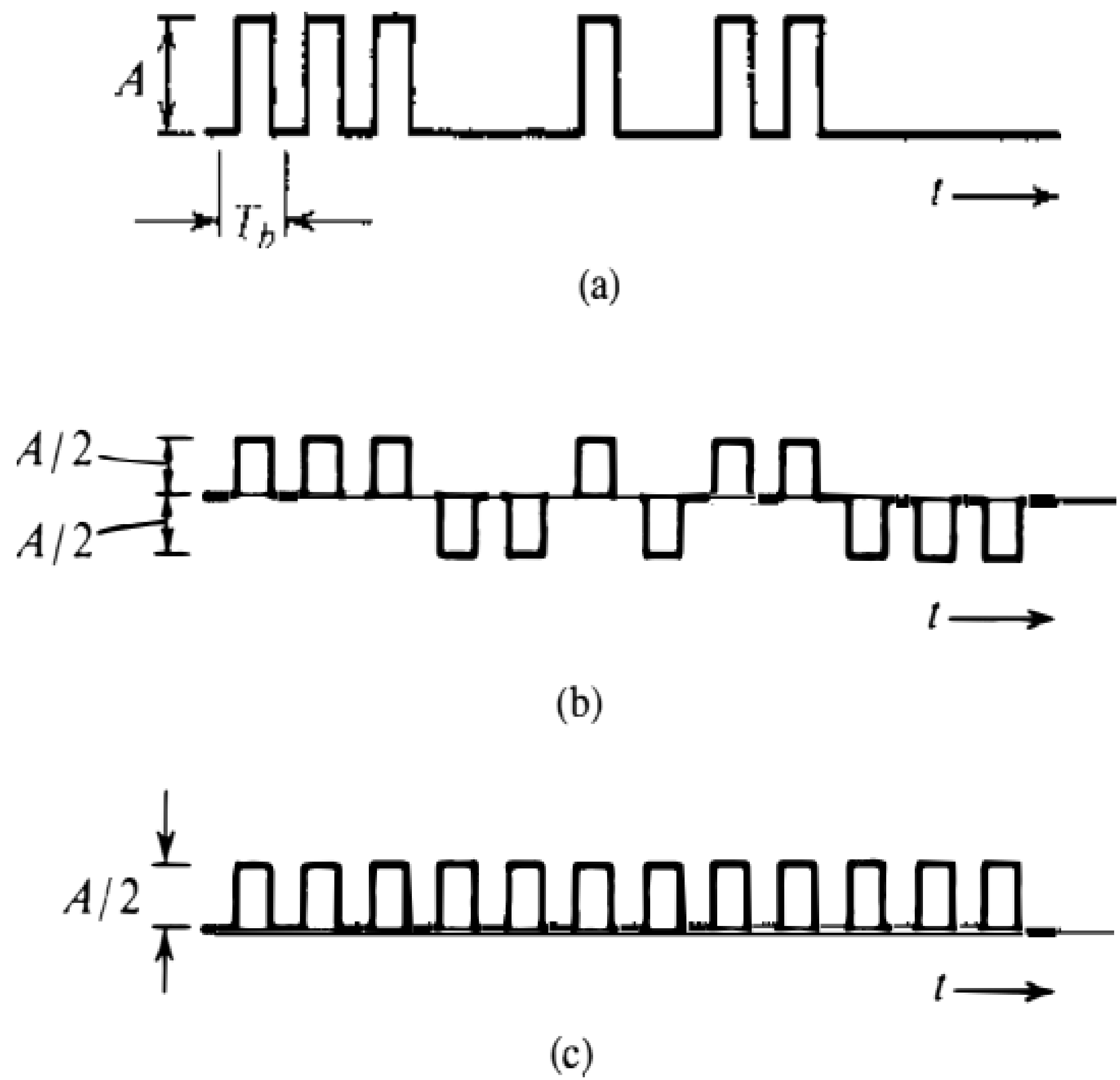
7. 1 . 4 Regenerative Repeater

Regenerative repeaters are used at regularly spaced intervals along a digital transmission line to detect the incoming digital signal and regenerate new "clean" pulses for further transmission along the line. This process periodically eliminates, and thereby combats, accumulation of noise and signal distortion along the transmission path. The ability of such regenerative repeaters to effectively eliminate noise and signal distortion effects is one of the biggest advantages of digital communication systems over their analog counterparts.

- If the pulses are transmitted at a rate of R_b pulses per second, we require the periodic timing information –the clock signal at R_b Hz– to sample the incoming pulses at a repeater or receiver.

- This timing information can be extracted from the received signal itself if the line code is chosen properly.
- When the RZ polar signal in Fig. 7.2b is rectified, for example, it results in a periodic signal of clock frequency R_b Hz, which contains the desired periodic timing signal of frequency R_b Hz.
- When this signal is applied to a resonant circuit tuned to frequency R_b, the output, which is a sinusoid of frequency R_b Hz, can be used for timing.
- The on-off signal can be expressed as a sum of a periodic signal (of clock frequency) and a polar, or random, signal as shown in Fig. 7.3. Because of the presence of the periodic component, we can extract the timing information from this signal by using a resonant circuit tuned to the clock frequency.

Figure 7.3
An on-off signal (a) is a sum of a random polar signal (b) and a clock frequency periodic signal (c).



- A bipolar signal, when rectified, becomes an on-off signal. Hence, its timing information can be extracted using the same way as that for an on-off signal.
- The timing signal (the resonant circuit output) is sensitive to the incoming bit pattern. In the on-off or bipolar case, a 0 is transmitted by 'no pulse.' Hence, if there are too many 0s in a sequence (no pulses), there is no signal at the input of the resonant circuit and the sinusoidal output of the resonant circuit starts decaying, thus causing error in the timing information.
- A line code in which the bit pattern does not affect the accuracy of the timing information is said to be a **transparent** line code. **The *RZ* polar** scheme (where each bit is transmitted by some pulse) is transparent, whereas the on-off and bipolar are nontransparent.

7.2 Line Coding

➢ **Desirable Properties for Line Codes**

- Transmission bandwidth: it should be as small as possible.
- Power efficiency: For a given bandwidth and a specified signal to noise ratio (SNR), the transmitted power should be as low as possible.
- Favorable power spectral density (PSD): It is desirable to have zero PSD at f = 0 (DC) and low PSD at low-frequency components because ac coupling and transformers are often used at the repeaters. Significant power in low-frequency components should also be avoided because it causes dc wander in the pulse stream when ac coupling is used.

- Error detection and correction capability: It is desirable to detect, and preferably correct, detected errors.
- *Adequate timing content.* It should be possible to extract timing or clock information from the signal (self synchronization).
- *Transparency.* It should be possible to correctly transmit a digital signal regardless of the pattern of 1s and 0s. We saw earlier that a long string of 0s could cause problems in timing extraction for the on-off and bipolar cases. A code is transparent if the data are so coded that for every possible sequence of data, the coded signal is received faithfully.

➢ Notes

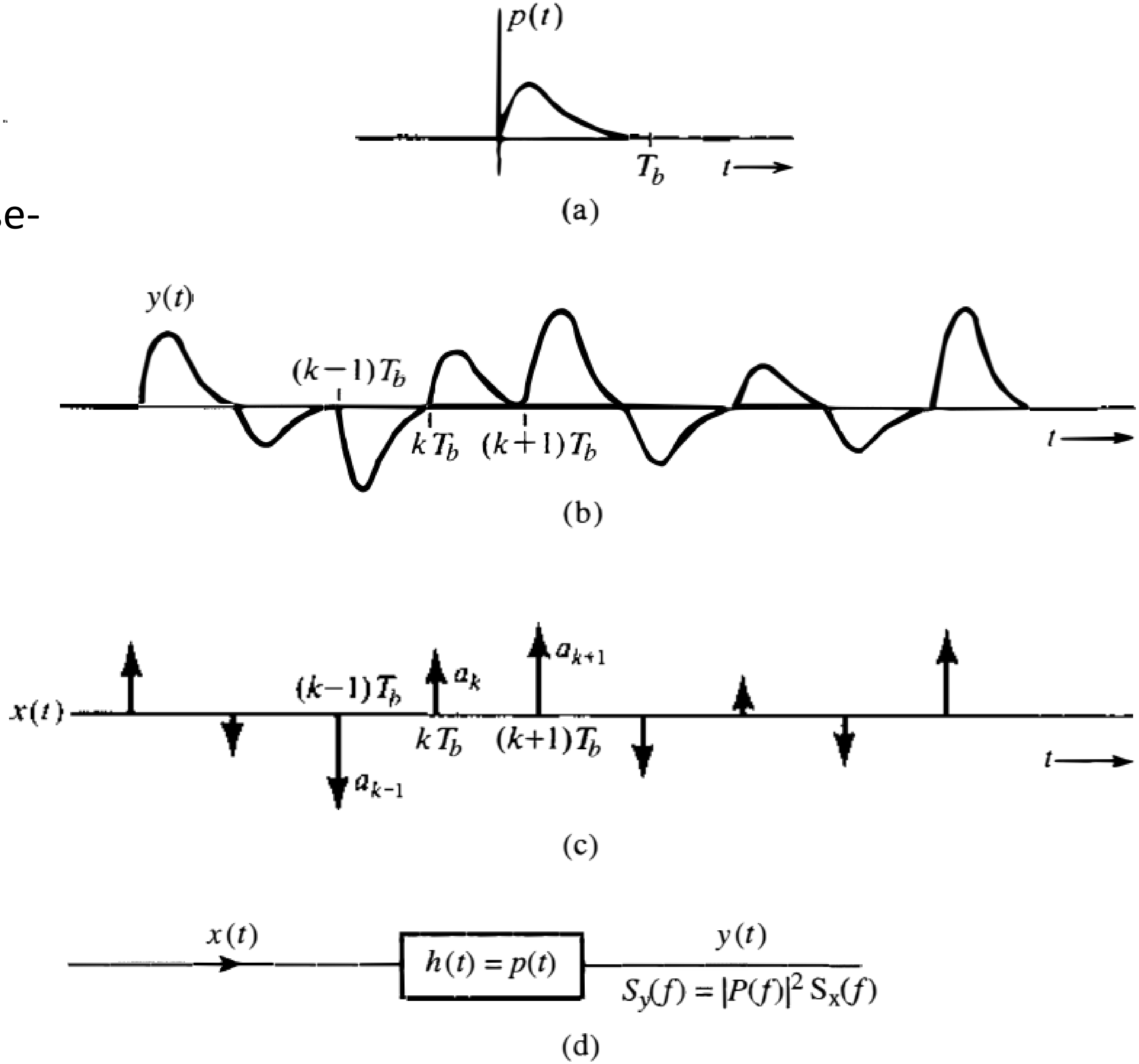
- The unipolar nature of an on-off signal results in a *DC component* that cause DC wander and wastes power as it carries no information.
- The polar code has opposite polarity pulses, so its DC component will be zero if the message contains 1s and 0s in equal proportion. In addition, it is the most power-efficient code because it requires the least power for a given noise immunity (error probability).
- The bipolar code also has zero DC component and it has the advantage that if *one single* error is made in the detecting of pulses, the received pulse sequence will violate the bipolar rule and the error can be detected.

7.2.1 PSD of Various Line Codes

- we assume that each transmitted bit is represented by a generic pulse p(t) whose corresponding Fourier transform is *P(f).* We can denote the line code symbol at time *k* as a_k .
- When the transmission rate is $R_b = 1/T_b$ pulses per second, the line code generates a pulse train constructed from the basic pulse *p(t)* with amplitude a_k starting at time $t = kT_b$; in other words, the k_{th} symbol is transmitted as $a_k p(t - kT_b)$.
- Figure 7.4a provides an illustration of a special pulse p(t), whereas Fig. 7.4b shows the corresponding pulse train generated by the line coder at baseband.
- As shown in Fig. 7.4b, counting a succession of symbol transmissions T_b second apart, the baseband signal is a pulse train of the form

$$y(t) = \sum a_k\, p(t - kT_b) \qquad (7.1)$$

Figure 7.4 Random pulse-amplitude modulated signal and its generation from a PAM impulse.



- The values a_k are random and depend on the line coder input and the line code itself. For example, for the on-off line code, $a_k = \{0, 1\}$, for the polar line code $a_k = \{1, -1\}$, and for the bipolar line code $a_k = \{0, 1, -1\}$, subject to some constraints.
- To determine the PSD for y(t) [$S_y(f)$], we use a signal *x(t)* that uses a unit impulse for the basic pulse *p(t)* (Fig. 7.4c). The impulses are at the intervals of T_b and the strength of the k_{th} impulse is a_k.
- So, if *x(t)* is applied to the input of a filter that has a unit impulse response *h(t)* = *p(t)* (Fig. 7.4d), the output will be the pulse train *y(t)* in Fig. 7.4b.
- Applying Eq. (3.92), the PSD of *y(t)* is

$$S_y(f) = |P(f)|^2 S_x(f) \qquad (7.2)$$

where $S_x(f)$ is the PSD of x(t).

- Recall that the PSD of a signal is the Fourier transform of its autocorrelation function. It can be shown that the autocorrelation of x(t) is

$$\mathcal{R}_x(\tau) = \frac{1}{T_b} \sum_{n=-\infty}^{\infty} R_n \delta(\tau - nT_b) \qquad (7.3)$$

where R_n is the discrete autocorrelation function of the line code symbols $\{a_k\}$ given by:

$$R_n = \lim_{T \to \infty} \frac{T_b}{T} \sum_k a_k a_{k+n} \qquad (7.4)$$

During the interval T, there are N pulses, so $T = NT_b$ this yields,

$$R_n = \lim_{N \to \infty} \frac{1}{N} \sum_k a_k a_{k+n} \qquad (7.5)$$

- R_n is the time average (mean) of the product $a_k a_{k+n}$ and in our notation is

$$R_n = \widetilde{a_k a_{k+n}}, \qquad n = 0, \pm 1, \pm 2, \cdots$$

- For example, for n = 0,

$$R_0 = \lim_{N \to \infty} \frac{1}{N} \sum_k a_k^2 = \widetilde{a_k^2}$$

- and for n = 1

$$R_1 = \lim_{N \to \infty} \frac{1}{N} \sum_k a_k a_{k+1} = \widetilde{a_k a_{k+1}}$$

- Since, the PSD $S_x(f)$ is the Fourier transform of $\mathcal{R}_x(\tau)$. Therefore,

$$S_x(f) = \frac{1}{T_b} \sum_{n=-\infty}^{\infty} R_n e^{-jn2\pi f T_b} \qquad (7.6)$$

- Recognizing that $R_{-n} = R_n$ [because $R(\tau)$ is an even function of τ], we have

$$S_x(f) = \frac{1}{T_b}\left[R_o + 2\sum_{n=1}^{\infty} R_n \cos n2\pi f T_b\right] \qquad (7.7)$$

- Thus, the PSD of a line code is fully characterized by its R_n and the pulse-shaping selection $P(f)$.
- This general result can be used to find the PSDs of various specific line codes by first determining the symbol autocorrelation R_n.